HIROTAKA (VICTOR) YIN

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Education

Carnegie Mellon University | Pittsburgh, PA — Expected December 2027

Master of Science in Mechanical Engineering | GPA: 3.99/4.0

Coursework: Mechatronics Design, Systems Dynamics, Embedded Computing, Computer Vision, Statistical Analysis

University of Washington | Seattle, WA — June 2025

Bachelor of Science in Mechanical Engineering | GPA: 3.84/4.0

Skills

Manufacturing: Design for Manufacturing, GD&T, Tolerance Analysis, Six-Sigma; CNC Machining, Plastic/Metal Part Design

Software: SolidWorks, OnShape, FEA, MATLAB, ROS/RViz, Microsoft Office, Python, Bash/Shell, Linux CLI, Git, MCP Frameworks

Languages: English, Japanese (Native), Chinese Mandarin (Professional Working Proficiency)

Work Experience

iPhone Product Design Intern | **Apple** | Shanghai, China; Shenzhen, China — March 2026 - August 2026

● Conducted systems-level failure analysis through fault injection and commonality analysis of Cpk and specification mean shifts; collaborated with upstream vendors and CMs to execute targeted material, fixture process and machine mitigation strategies, boosting process capability and units-per-hour (UPH) while capping part reject rates below 0.02%.

● Engineered an end-to-end pipeline using deep learning to automate defect detection, outperforming human inspection reliability and helping address a primary UPH bottleneck. Presented data-driven reports and helped initiate inline factory integration.

● Assisted development of a multi-agentic platform used by in-region teams by leveraging LLM agents across manufacturing enterprise databases, expanding analysis capabilities and cutting response latency by up to 50% to deliver smarter, faster decisions.

Research Experience

Autonomous Drones Research |CERLAB, Carnegie Mellon University — August 2025 - Present

● Developed and validated 3D reinforcement learning-based local planners for autonomous UAVs using a Sim2Real workflow with ROS, Gazebo and RViz, resolving computational bottlenecks of previous roadmap-based algorithms when exploring large environments, delivering up to a 50% improvement in trajectory generation time.

● Designed custom LiDAR attachments in SolidWorks, expanding below-horizon FOV by 30%, improving collision avoidance during descent. Optimized airframe through FEA for stress concentrations and vibration modes, improving flight stability.

Laboratory Robotics Research | Sun Lab, University of Washington — October 2023 - August 2025

● Eliminated manual sample inspection by designing an open-source, ~$160 two-axis spectrometry gantry for a liquid-handling robot, with substrate fixtures and precision probe alignment, increasing throughput by 4x and enabling closed-loop automation.

● Built a closed-loop Bayesian optimization workflow linking automated pipetting, mixing and substrate inspection, replacing previously trial-and-error parameter tuning; demonstrated a working prototype for accelerated solar-cell material optimization.

● Co-authored and presented research on AI-driven laboratory automation workflows (DOI: 10.5206/joh.v9i1.23044).

Projects

Underwater Inspection Robot | *Soft-Robotics Project,* Carnegie Mellon University — October 2025 - December 2025

● Designed novel actuation mechanisms for a bio-inspired stingray robot by combining soft materials with Shape Memory Alloy (SMA) actuators, improving propulsion efficiency by 15% and minimizing sediment disturbance for applications in marine monitoring.

● Engineered TPU seals and multi-stage silicone molding processes to mitigate fluid intrusion into SMA wires, leveraging deformation FEA to optimize seal compression and waterproofing reliability without compromising ease-of-assembly.

Autonomous Cleaning Robot | *Senior Capstone Project,* University of Washington — January 2025 - June 2025

● Conducted Kano analysis to identify key subsystem requirements and designed modular snap-fit chassis assemblies, enabling rapid subsystem-prototyping, streamlining design down-selection and cutting development time by 2 weeks.

● Engineered high-torque drivetrains and built MATLAB simulations to validate navigation capabilities, improving speed, load capacity and traversal reliability to meet customer needs, while optimizing for assembly and fabrication across a team of 5.